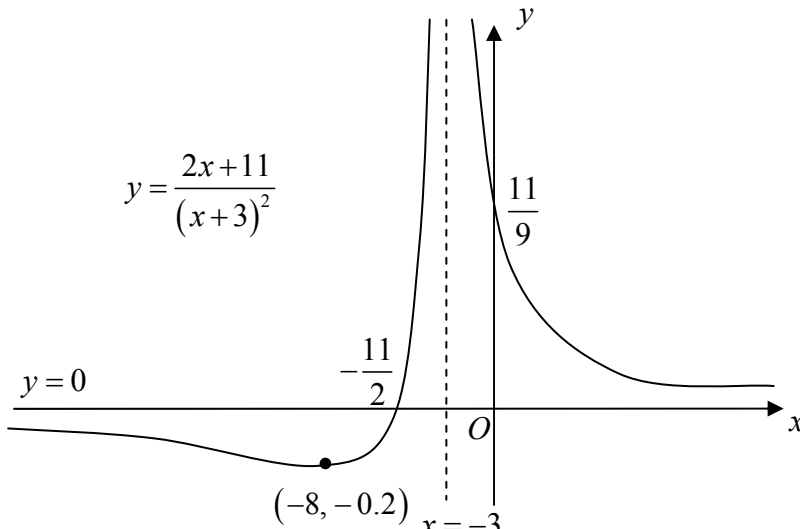
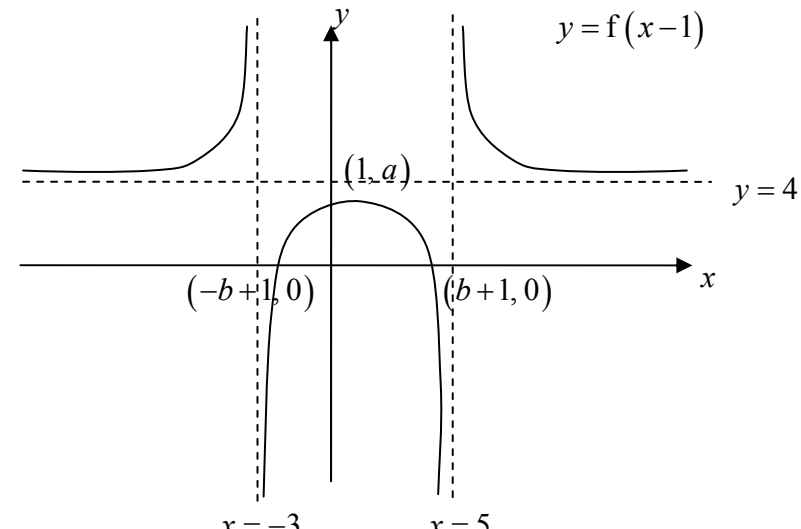
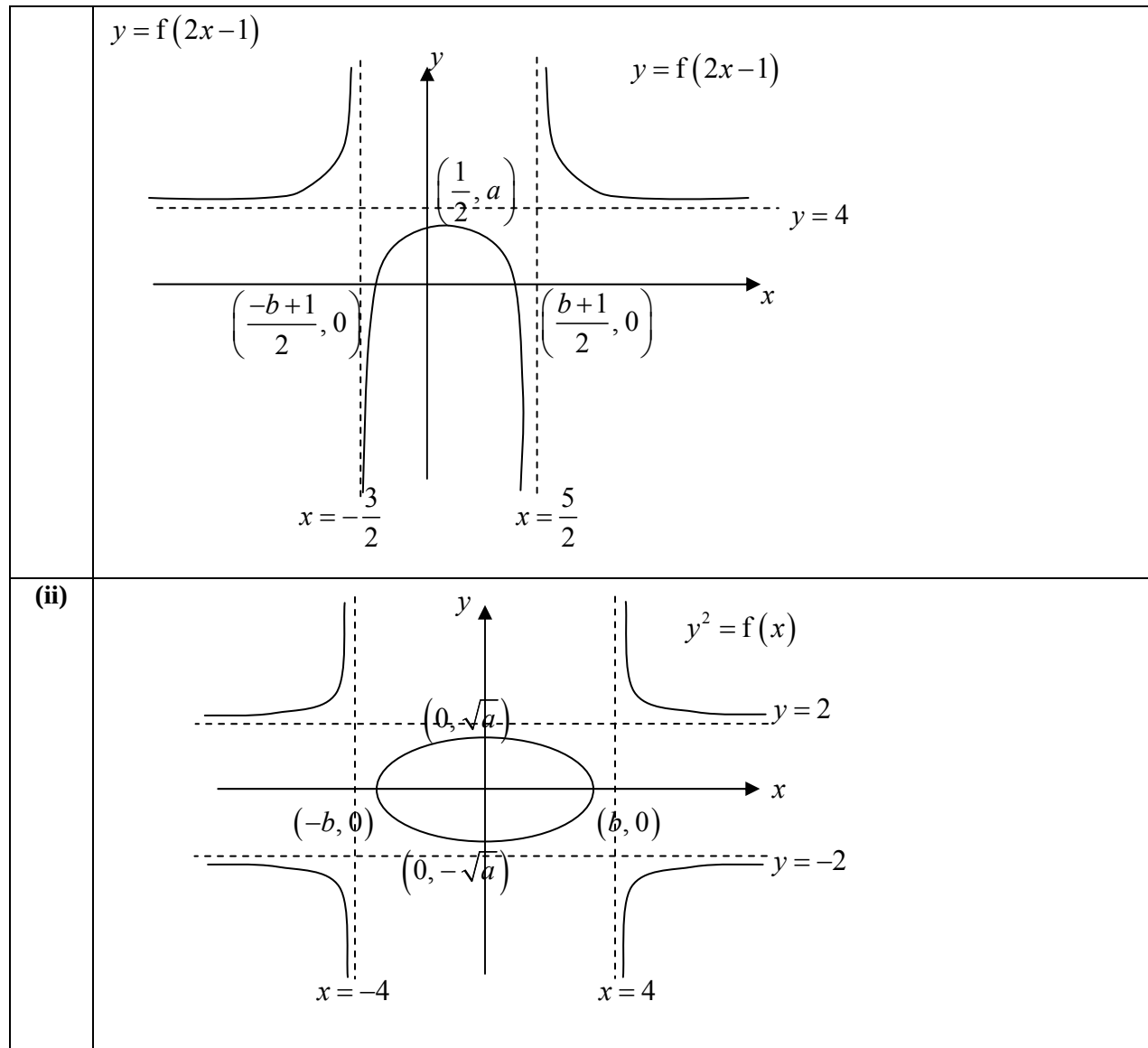


2009 MJC JC2 Preliminary Exam
H2 Mathematics (9740) Paper 2 Suggested Solution

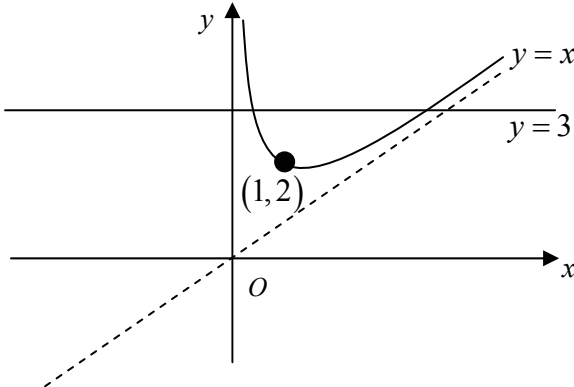
Qn	Solution
1	Integration by parts
(i)	$\frac{d}{dx} e^{\cos x} = -e^{\cos x} \sin x.$
(ii)	$\int e^{\cos x} \sin 2x \, dx$ $= \int e^{\cos x} (2 \sin x \cos x) \, dx$ $= \int (-e^{\cos x} \sin x)(-2 \cos x) \, dx$ $= -2e^{\cos x} \cos x - 2 \int e^{\cos x} \sin x \, dx$ $= -2e^{\cos x} \cos x + 2e^{\cos x} + C$

Qn	Solution
2	Graphs
(a)	 Graph of the function $y = \frac{2x+11}{(x+3)^2}$. The graph shows a vertical asymptote at $x = -3$ and a horizontal asymptote at $y = 0$. The curve passes through the point $(-8, -0.2)$ and the point $(0, \frac{11}{9})$. The origin is labeled O.
(b)(i)	 Graph of the function $y = f(x-1)$. The graph shows vertical asymptotes at $x = -3$ and $x = 5$, and a horizontal asymptote at $y = 4$. The curve passes through the points $(-b+1, 0)$, $(1, a)$, and $(b+1, 0)$.



Qn	Solution
3	Equations (System of equations) and Inequalities
(a)	$y = a\sqrt{x} + \frac{b}{x} + cx^2 - d$ $\frac{dy}{dx} = \frac{a}{2\sqrt{x}} - \frac{b}{x^2} + 2cx$ Augmented matrix $\left(\begin{array}{cccc c} 1 & 1 & 1 & -1 & -9 \\ 0.5 & -1 & 2 & 0 & 0 \\ 2 & \frac{1}{4} & 16 & -1 & \frac{23}{4} \\ 3 & \frac{1}{9} & 81 & -1 & \frac{217}{3} \end{array} \right) \quad \text{OR} \quad \begin{array}{l} a + b + c - d = -9 \\ \frac{1}{2}a - b + 2c = 0 \\ 2a + \frac{1}{4}b + 16c - d = \frac{23}{4} \\ 3a + \frac{1}{9}b + 81c - d = \frac{217}{3} \end{array}$ Using GC, $a = 2, \quad b = 3 \quad c = 1 \quad d = 15$ The equation of the curve is $y = 2\sqrt{x} + \frac{3}{x} + x^2 - 15$
(b)	$4x^2 + 4x + 3 = (2x+1)^2 + 2 > 0 \quad \forall x \in \mathbb{R}$ Alternative Method Discriminant $= 4^2 - 4(4)(3) = -32$ Since coefficient of x^2 is positive and discriminant < 0, $4x^2 + 4x + 3$ is positive
(ii)	$\frac{4x^3 + 4x^2 + 3x}{x^2 + x - 2} < 0$ $\frac{x(4x^2 + 4x + 3)}{(x+2)(x-1)} < 0$ $\frac{x}{(x+2)(x-1)} < 0 \quad \text{since } (2x+1)^2 + 2 > 0 \quad \forall x \in \mathbb{R}$ $\begin{array}{ccccccc} & - & & + & & - & & + \\ & & & & & & & \\ & -2 & & 0 & & 1 & & \end{array}$ Therefore $x < -2$ or $0 < x < 1$

$$0 < x < e^{-1} \quad \text{or} \quad x > e$$

Qn	Solution
4	Functions
(i)	 <u>Either:</u> Any line $y = k, k > 2$ cuts the graph twice; hence f is not 1-1 and f^{-1} does not exist. <u>OR:</u> $y = 3$ cuts the graph twice; hence f is not 1-1 and f^{-1} does not exist. The largest value of k is 1.

Let $y = x + \frac{1}{x}$,

$$x^2 - xy + 1 = 0$$

$$x = \frac{y \pm \sqrt{y^2 - 4}}{2}$$

Reject $x = \frac{y + \sqrt{y^2 - 4}}{2} \because 0 < x \leq 1$.

$$\therefore x = \frac{y - \sqrt{y^2 - 4}}{2}$$

$$f^{-1}(x) = \frac{x - \sqrt{x^2 - 4}}{2}$$

$$D_{f^{-1}} = R_f = [2, \infty)$$

$$f^{-1}: x \mapsto \frac{x - \sqrt{x^2 - 4}}{2}, \quad x \geq 2.$$

For hg to exist, $R_g \subseteq D_h$.

$$R_g = (\lambda - 9, \lambda], \quad D_h = (0, \infty).$$

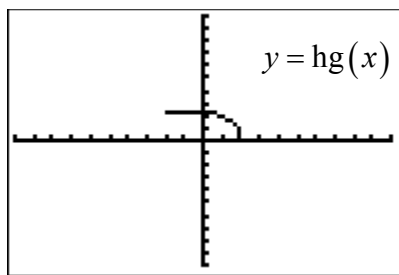
Hence, the minimum value of λ is 9.

To find the R_{hg} ,

Method 1

$$D_g = (-2, 2) \rightarrow R_g = (0, 9] = D_h \rightarrow R_{hg} = (\ln 2, \ln 11]$$

Method 2



From the graph,

$$R_{hg} = (\ln 2, \ln 11].$$

Qn	Solution
5	Integration – Area and Volume
(a)	$x^2 + (y - a)^2 = a^2$ $\Rightarrow x^2 = a^2 - (y - a)^2$ Volume formed $= \pi \int_0^{2a} [a^2 - (y - a)^2] dy$ $= \pi \int_0^{2a} [a^2 - (y^2 - 2ay + a^2)] dy \quad \text{OR} = \pi \left[a^2 y - \frac{1}{3} (y - a)^3 \right]_0^{2a}$ $= \pi \int_0^{2a} [a^2 - (y^2 - 2ay + a^2)] dy = \pi \left[\left(2a^3 - \frac{1}{3} a^3 \right) - \left(0 - \frac{1}{3} (-a^3) \right) \right]$ $= \pi \int_0^{2a} [-y^2 + 2ay] dy = \frac{4}{3} \pi a^3$ $= \pi \left[-\frac{y^3}{3} + ay^2 \right]_0^{2a}$ $= \pi \left[-\frac{8a^3}{3} + 4a^3 \right]$ $= \frac{4}{3} \pi a^3$ Volume of sphere with radius a is $= \frac{4}{3} \pi a^3$ Therefore volume of the semi-circle obtained when rotated 2π radian about the y-axis is equal to the volume of a sphere with radius a
	When $x = \ln\left(\frac{\sqrt{3}}{2}\right) \Rightarrow t = 2$ $x = \ln\left(\frac{\sqrt{24}}{5}\right) \Rightarrow t = 5$ $x = \ln \frac{(t^2 - 1)^{\frac{1}{2}}}{t} = \frac{1}{2} \ln(t^2 - 1) - \ln t$

	$\frac{dx}{dt} = \frac{1}{2} \left(\frac{2t}{t^2 - 1} \right) - \frac{1}{t}$ $\frac{dx}{dt} = \frac{1}{t(t^2 - 1)}$ $\text{Area of required region} = \int_{\ln \frac{\sqrt{3}}{2}}^{\ln \frac{\sqrt{24}}{5}} y \, dx$ $= \int_2^5 y \frac{dx}{dt} \, dt$ $= \int_2^5 t(5t^2 - 8) \times \frac{1}{t(t^2 - 1)} \, dt$ $= \int_2^5 \frac{5t^2 - 8}{(t^2 - 1)} \, dt$ $= \int_2^5 5 - \frac{3}{(t^2 - 1)} \, dt$ $= \left[5t - \frac{3}{2} \ln \left(\frac{t-1}{t+1} \right) \right]_2^5$ $= \left[25 - \frac{3}{2} \ln \frac{4}{6} \right] - \left[10 - \frac{3}{2} \ln \frac{1}{3} \right]$ $= 15 + \frac{3}{2} \ln \frac{1}{2} \text{ OR } = 15 - \frac{3}{2} \ln 2$
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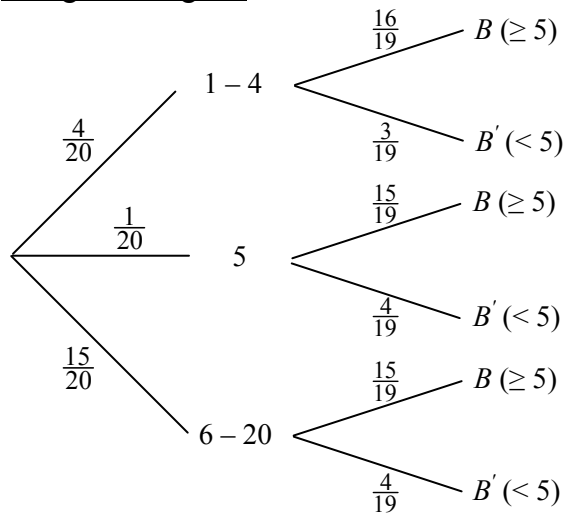
Qn	Solution
6	Sampling Methods
	A sample is a small representative of the population. In many practical statistical investigations, it may not be possible to collect information for the entire population due to time or cost constraints (or any other sensible reasons). Hence it is desirable to use the results obtained from a sample to make statistical inferences about the population. Any sensible real-life example which gives a complete sampling frame and ensures that every member of the population has an equal chance of being selected.

Qn	Solution
7	Hypothesis Testing
	Unbiased estimate of population variance = $\frac{80}{79}(11.23)^2$ $= 127.71$ $H_0: \mu = 1000$ $H_1: \mu \neq 1000$ Since $n = 80$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(\mu, \frac{s^2}{n}\right) \text{ approximately}$ Test statistic : $Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$ Level of significance: 4% Critical Region: Reject H_0 if $p\text{-value} < 0.04$ Assuming H_0 is true, from the graphic calculator, $p\text{-value} = 0.0687$ Conclusion: Since $p\text{-value} = 0.0687 > 0.04$, we do not reject H_0 and conclude that there is no significant evidence, at 4% level, the mean volume is not 1000 ml.

Qn	Solution
8	Permutation and Combination
(a)(i)	1 'B', 2 'E', 1 'A', 3 'N', 3 'T' No. of ways = $\frac{8!}{3!2!2!} = 1680$
(ii)	No. of ways = $\frac{4!}{2!} \times \binom{5}{3} \times \binom{8}{3} = 6720$
(b)(i)	No. of ways = No. of ways = $\binom{5}{4} \times \binom{10}{4} 8! = 42336000$
(ii)	No. of ways = $\binom{3}{1} \binom{4}{2} \binom{10}{1} \binom{9}{2} \frac{8!}{2!2!} = 65318400$

Qn	Solution
9	Probability
(a)(i)	P(all three cards even) = $\frac{10}{20} \frac{9}{19} \frac{8}{18} = \frac{2}{19}$ or $= \frac{\binom{10}{3}}{\binom{20}{3}} = \frac{2}{19}$
(ii)	P(exactly one even) = $\frac{10}{20} \frac{10}{19} \frac{9}{18} \times \binom{3}{1} = \frac{15}{38}$ or $= \frac{\binom{10}{1} \binom{10}{2}}{\binom{20}{3}} = \frac{15}{38}$
(b)	$P(B) = \frac{4}{20} \frac{16}{19} + \frac{16}{20} \frac{15}{19} = \frac{4}{5}$ $P(A \cap B) = P(\text{1st card} \leq 5 \text{ \& } 2\text{nd card} \geq 5)$ $= P(\text{1st card} = 5 \text{ \& } 2\text{nd card} > 5) + P(\text{1st card} < 5 \text{ \& } 2\text{nd card} \geq 5)$ $= \frac{1}{20} \frac{15}{19} + \frac{4}{20} \frac{16}{19}$ $= \frac{79}{380}$ $P(B A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{79}{380}}{\frac{1}{4}} = \frac{79}{95}$

Using tree diagram:



(i) $P(B) = \frac{4}{20} \frac{16}{19} + \frac{1}{20} \frac{15}{19} + \frac{15}{20} \frac{15}{19} = \frac{4}{5}$

(ii) $P(A \cap B) = \frac{4}{20} \frac{16}{19} + \frac{1}{20} \frac{15}{19} = \frac{79}{380}$

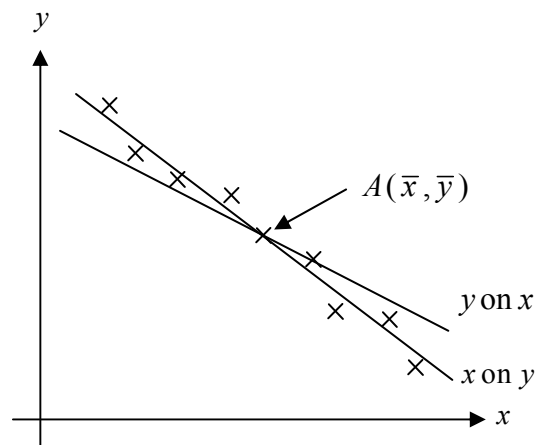
(iii) $P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{79}{380}}{\frac{1}{4}} = \frac{79}{95}$

Qn	Solution
10	Binomial and Poisson Distributions (include approximation)
(i)	Let C and P be the number of cheese and peanut waffles sold in a randomly chosen 15-minute period respectively. $C \sim P_o(4)$, $P \sim P_o(5)$ $C + P \sim P_o(4 + 5) \Rightarrow P_o(9)$ $P(C + P \geq 10) = 1 - P(C + P \leq 9)$ $= 0.41259$ ≈ 0.413
(ii)	Required probability = $P(\text{all waffles sold are peanut flavour} \mid \text{at least 10 waffles are sold})$ $= \frac{P(\text{all waffles sold are peanut flavour and at least 10 waffles are sold})}{P(\text{at least 10 waffles are sold})}$ $= \frac{P(P \geq 10) \cdot P(C = 0)}{P(C + P \geq 10)}$ $= \frac{[1 - P(P \leq 9)] \cdot P(C = 0)}{P(C + P \geq 10)}$ $= \frac{(0.031828) \cdot (0.018316)}{0.41259}$ $= 0.00141$
(iii)	$P(\text{at least 10 waffles are sold in each of the 4 consecutive 15-minute period in a randomly selected hour}) = (0.41259)^4 \text{ or } (0.413)^4$ $= 0.0290 \text{ or } 0.0291 \text{ (to 3 s.f.)}$ Alternatively Let T be the number of 15-minute periods in a randomly chosen one-hour period where at least 10 waffles are sold. $T \sim B(4, 0.41259)$ Required probability = $P(T = 4) = 0.0290 \text{ or } 0.0291 \text{ (to 3 s.f.)}$
(iv)	Let X be the number of waffles sold in an hour. $X \sim P_o(9 \times 4) \Rightarrow X \sim P_o(36)$ Since $\lambda = 36 > 10$, $X \sim N(36, 36)$ approximately $P(X \geq 40) = P(X > 39.5)$ after continuity correction $= 0.280 \text{ (to 3 s.f.)}$ Let A be the event that at least 10 waffles are sold in each of the 15 minute period in an hour. Let B be the event that at least 40 waffles are sold in an hour. Obviously, event A is a subset of B and hence the probability of A is smaller than that of B

Qn	Solution
11	Correlation and Regression
(a)(i)	$r = 0.89241 = 0.892$ (3 s.f.)
(a)(ii)	$r = 0.95956 = 0.960$ (3 s.f.) Since $r = 0.960$ is closer to 1 than $r = 0.892$, the model in (i) is less suitable than the model in (ii).
(a) (iii)	Regression line of F on x : $F = 0.35903 + 0.029245x$ $F = 0.359 + 0.0292x$ Regression line of x on F : $x = 204.51 + 31.484F$ $x = 205 + 31.5F$
(a) (iv)	Using $F = 0.35903 + 0.029245x$, $100 = 0.35903 + 0.029245t^2$ $t = 58.37047 = 58.4$ s (3 s.f.)
(b) (i)	Regression line of x on y : $3x + 2y = 51 \Rightarrow x = \frac{51}{3} - \frac{2}{3}y$ Regression line of y on x : $8x - 11y = 36 \Rightarrow y = -\frac{36}{11} + \frac{8}{11}x$ Since $bd = \left(\frac{8}{11}\right)\left(-\frac{2}{3}\right)$ is negative and r^2 is always positive, therefore $bd \neq r^2$. Hence $8x - 11y = 36$ is not valid. OR Since b and d must be both positive or both negative, but $b = \left(\frac{8}{11}\right)$ and $d = \left(-\frac{2}{3}\right)$, hence $8x - 11y = 36$ is not valid.

(b)

(ii)



From $x = \frac{51}{3} - \frac{2}{3}y$, we have $d = -\frac{2}{3}$.

Gradient of regression line of x on y in graph of y against x is $\frac{1}{d} = \frac{1}{-\frac{2}{3}} = -\frac{3}{2}$

Since $r = -0.943$,

Using $bd = r^2$,

$$b\left(-\frac{2}{3}\right) = (-0.943)^2$$

$$b = -1.33 > -\frac{3}{2}$$

Therefore regression line y on x has more gentle slope than regression line x on y .

Qn	Solution
12	Normal Distribution (include Sample Mean CLT)
(a)	By symmetry, $\mu = \frac{155+185}{2} = 170$ $P(X < 155) = 0.025$ $\Rightarrow P\left(Z < \frac{155-170}{\sigma}\right) = 0.025$ $\Rightarrow -\frac{15}{\sigma} = -1.95996$ $\Rightarrow \sigma = 7.6532$ $\therefore \sigma^2 = 58.6$ (3 s.f.)
(i)	$X_1 + X_2 - 2X_3 \sim N(0, 6\sigma^2)$ $E(X_1 + X_2 - 2X_3) = E(X) + E(X) - 2E(X) = 0$ $\text{Var}(X_1 + X_2 - 2X_3) = 2\text{Var}(X) + 4\text{Var}(X) = 6\sigma^2$ $P(X_1 + X_2 > 2X_3 + 5)$ $= P(X_1 + X_2 - 2X_3 > 5)$ $= 0.39484$ $= 0.394$ (3 s.f.)
(ii)	$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{50}\right)$ (Since X is normally distributed, there isn't a need to use CLT.) $P(\bar{X} < 172)$ $= 0.96769$ $= 0.968$ (3 s.f.)
(b)	Since n is large, by CLT $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approx.

$$P\left(\left|\bar{X} - \mu\right| < 1\right) > 0.99$$

$$\Rightarrow P\left(\left|Z\right| < \frac{1}{5/\sqrt{n}}\right) > 0.99$$

$$\Rightarrow P\left(-\frac{1}{5/\sqrt{n}} < Z < \frac{1}{5/\sqrt{n}}\right) > 0.99$$

$$\Rightarrow 1 - 2P\left(Z < -\frac{1}{5/\sqrt{n}}\right) > 0.99$$

$$\Rightarrow P\left(Z < -\frac{1}{5/\sqrt{n}}\right) < 0.005$$

$$\Rightarrow -0.2\sqrt{n} < -2.5758$$

$$\Rightarrow 0.2\sqrt{n} > 2.5758$$

$$\Rightarrow n > 165.87$$

Least $n = 166$.